

Health Care Benefits Test Plan

In this test plan, I will cover all aspects of testing from the initial User Stories creation to Risk Assessment. Most of the techniques used draw their inspiration from our textbook (Van Veenendaal et al., 2019a).

User Stories

User Story 1 - View Current Benefit Selections

As an employee, I need to view my current medical, dental, and vision benefit selections and my covered dependents, so that I can verify what benefits coverage my family and I will have.

Acceptance Criteria

- The application displays all selected benefit plans for the employee.
- All covered dependents are listed under each applicable plan.
- The plan type of PPO or HMO is clearly identified.
- If the employee has no benefit selections, the system clearly displays a “no benefits are selected” message.

User Story 2 - Modify Benefit Selections During the Open Enrollment

As an employee, I need to change my benefit selections during the open enrollment, so that I can customize my benefits coverage.

Acceptance Criteria

- The system allows benefit plan changes only during the enrollment period.
- The employee can switch between PPO and HMO for health care coverage.
- Optional dental and vision coverage can be added or removed.
- Changes are saved immediately in the benefits summary which is displayed.
- The system saves all changes successfully.

User Story 3 - Add or Remove Dependents

As an employee, I need to add or remove dependents for each benefit coverage type, so that my family is covered correctly.

Acceptance Criteria

- The system allows entry of new dependent names, their relationship, and all required personal information.
- Dependents can be removed from the benefits coverage selections.
- Updates immediately adjust the benefits coverage summary.
- The system prevents duplicate dependent entries.
- All changes are saved on the employee benefits record.

User Story 4 - Require Employee Coverage Before Dependent Coverage

As an employee, I must enroll myself in a benefits coverage before enrolling a dependent in the same benefits coverage, so that the benefits coverage rules are being enforced.

Acceptance Criteria

- The system prevents selecting dependent coverage if employee benefits coverage is not yet selected.
- A clear validation message explains this benefits requirement.
- Invalid selections cannot be saved.
- Once employee benefits coverage is selected, dependent benefits coverage becomes available.

User Story 5 - Determine Coverage Tier Automatically

As an employee, I need the system to apply the correct benefits coverage tier, so that my benefits costs are calculated correctly.

Acceptance Criteria

- The system assigns the correct coverage tier based on the selected dependents.
- The coverage tier is updated automatically when dependents are added or removed.
- All four of the coverage tiers are available (employee only, employee and spouse, employee, spouse and one child, employee and family).
- The assigned coverage tier is visible in the summary view.

User Story 6 - Benefits Enrollment Form

As an employee, I need to receive a benefits enrollment form, so that I can review and sign up for my benefits selections.

Acceptance Criteria

- The system generates a form reflecting the current benefits selections.
- The form includes the employee and dependent information.
- The form cannot be submitted if there is required information that is missing.
- The submitted forms can be viewed or printed.

User Story 7 - Update Effective Date

As a Human Resources employee, I need to update the effective date, so that benefits begin on the correct date.

Acceptance Criteria

- Only a Human Resources employee can edit the effective date.
- The system validates the entered date.
- The updated date appears on the enrollment form.
- All changes are recorded in the system.

User Story 8 - Maintain Employee Data

As a Human Resources employee, I need to maintain employee records, so that employee information stays accurate.

Acceptance Criteria

- Required fields include employee ID, department ID, name, address information, and office extension number.
- The system validates required fields.
- Duplicate employee IDs are not allowed.
- Updates are saved and are visible immediately.

User Story 9 - Maintain Insurance Company Data

As a Human Resources employee, I need to maintain insurance provider information, so that available plans are tracked correctly.

Acceptance Criteria

- Insurance company numbers and company names can be added or edited.
- The system prevents duplicating company numbers.
- Existing enrollment records are not affected by updates.
- Insurance providers can be marked as inactive without being deleted.

User Story 10 - Maintaining Rates Tables

As a Human Resources employee, I need to maintain benefit rates, so that benefits cost calculations are accurate.

Acceptance Criteria

- Rates can be defined for each plan and tier combination.
- Invalid rate values are rejected.
- Updates apply only to the effected period.
- Changes made are saved in the new calculations.

User Story 11 - Display Cost Responsibilities

As an employee, I need to see which costs are company-paid and which I must pay, so that I understand how much I must pay for benefits.

Acceptance Criteria

- Medical costs which apply are displayed as company paid.
- Dental and vision benefits costs are displayed as employee paid.
- A clear benefits cost breakdown is shown.
- Totals are updated when benefits selections have been changed.

User Story 12 - Prevent Form Submission When the Data is Incomplete

As an employee, I need the system to block the form submission when required information is still missing, so that errors in my requested benefits are prevented.

Acceptance Criteria

- The system validates all of the required data before allowing form submission.
- Input fields missing information are highlighted.
- A message clearly explains what must be corrected before the form can be submitted.
- No form is submitted until validation has passed.

User Story 13 - Remove My Coverage Removes My Dependents Coverage

As an employee, I need my dependents coverage to be removed when I remove my own coverage, so that invalid selections are not remaining in my benefits selections.

Acceptance Criteria

- Removing an employee's coverage automatically removes their dependents coverage.
- Benefits cost totals are recalculated immediately.
- The system prevents any orphaned dependents benefits selections.
- The updated benefits summary reflects the changes made.

User Story 14 - Save Benefits Selections Draft

As an employee, I need to save unfinished benefits selections, so that I can return and complete them later.

Acceptance Criteria

- The system allows the saving of incomplete benefits selections as a draft.
- Draft data is restored when the employee returns.
- Drafts do not affect active benefits enrollment until they have been submitted.
- Employees can continue editing a saved draft.

User Story 15 - Confirmation and Submission of Benefits Selections

As an employee, I need to confirm my benefits selections before they are submitted, so that benefits enrollment mistakes are minimized.

Acceptance Criteria

- A confirmation screen summarizes all benefit selections.
- The employee must approve the benefits selections before submission occurs.
- Submitted benefits selections are prevented from any further editing.
- The submission of the selected benefits is recorded successfully.
- The employee is notified on the display that the submission was successfully recorded.

User Story 16 - Searching and Viewing of Employee Records

As a Human Resources employee, I need to be able to search employee's benefits enrollment records, so that I can assist employees.

Acceptance Criteria

- Records can be searched by name or employee ID.
- The employees' current and past benefits enrollment data is viewable.
- Only an authorized Human Resources employee can access benefits records.
- The search results display accurate and up to date employee benefits information.

User Story 17 - Managing Dependents Records

As a Human Resources employee, I need to edit dependent information, so that records stay accurate.

Acceptance Criteria

- Dependents can be edited or deactivated without being deleted.
- A dependent relationship type must be specified.
- The changes saved are updated in the system.
- A history of data changes is stored.

User Story 18 - Maintain Department Information

As a Human Resources employee, I need to maintain department data, so that employee department assignments stay valid.

Acceptance Criteria

- Departments can be added or updated.
- Invalid departments cannot be assigned to employees.
- Departments which are in use cannot be deleted.
- Updates are saved immediately.

User Story 19 - Inactivate Plan Options

As a Human Resources employee, I need to switch plans to inactivate without deleting its history, so that past enrollments history stays intact.

Acceptance Criteria

- Inactive plans cannot be selected by employees.
- Plan enrollments history stays accessible.
- The system clearly marks a plan as inactive.

User Story 20 - Calculate Benefits Costs

As the system, I need to calculate benefits costs based on selected plans and tiers, so that benefits cost totals are accurate.

Acceptance Criteria

- Benefits costs are calculated using the correct rate table.
- Cost calculations are updated when the selections have changed.
- Benefits cost totals match defined plan and tier rates.
- Any invalid plan and tier combinations are rejected.

User Story 21 - Apply the Effective Date Amounts

As a Human Resources employee, I need rates defined for effective periods, so that the correct rates are being used.

Acceptance Criteria

- Benefits rates are associated with the defined period.
- The system selects proper rates based on the enrollments effective date.
- Any expired benefit rates are not being used.
- Future rates are applied when they take effect.

User Story 22 - Display Plan and Tier Cost Breakdowns

As an employee, I need to see detailed plan/tier cost summaries, so that I can compare my plan and tier benefits options.

Acceptance Criteria

- The Plan and Tier costs are displayed by coverage type.
- A total monthly cost is shown.
- Any changes made will update the summary of costs immediately.
- The information on costs is clear and readable.

User Story 23 - Enforcing The Benefits Enrollment Period

As the system, I need to restrict benefits changes to the allowable enrollment period, so that plan enrollment rules are being followed.

Acceptance Criteria

- Benefit selections can only be altered during the allowed enrollment period.
- Outside of the enrollment period, the benefits information is read-only.
- The system displays a message if editing of the benefits outside of the enrollment period is attempted.
- Authorized Human Resources employee may override the outside of enrollment period lock out if necessary.

User Story 24 - Validate the Employee's Data

As the system, I need to validate an employee's information, so that records stay accurate.

Acceptance Criteria

- Required data fields must be completed.
- Data field formats are being validated.
- Duplicate employee records are being prevented.
- Any errors are being clearly communicated.

User Story 25 - Validate Employee Dependent Relationships

As the system, I need valid the employee dependent relationships, so that benefits coverage eligibility rules are being enforced.

Acceptance Criteria

- A relationship type must be selected for all dependents.
- Duplicates of a dependent are not allowed.
- All invalid entries are being rejected.

User Story 26 - Handling Coverage Tier Changes

As the system, I need to update the coverage tier pricing when dependents are added or removed, so that benefits pricing stays accurate.

Acceptance Criteria

- The coverage tier automatically recalculates the total benefits costs after dependent changes.
- The correct coverage tier is always displayed based on the dependents included.
- The benefits costs are adjusted accordingly.

User Story 27 - Track Changes

As a Human Resources employee, I need an auditable trail of benefits changes, so that benefits changes can be reviewed.

Acceptance Criteria

- The system records who made each benefits change.
- The date and time of the changes are stored.
- Any modified fields are identifiable.
- Auditable data is read only.

User Story 28 - Clear Error Messages

As an employee, I need understandable error messages, so that I can correct any issues quickly.

Acceptance Criteria

- Error messages describe the exact issue.
- The error messages appear near the field which is affected.
- Instructions for how to correct the issue are provided.
- The system prevents submission without first correcting the invalid entry.

User Story 29 - Authenticate Users

As a user, I need to log in securely, so that my personal information is protected.

Acceptance Criteria

- The user must provide valid credentials.
- All invalid login attempts are rejected.
- Access to the system is only granted after authentication.
- A session only begins after login is successful.

User Story 30 - Enforce Role-Based Access

As the system, I need to restrict functions by role, so that users only perform authorized actions.

Acceptance Criteria

- Regular employees cannot access administrative functions.

- Human Resources employees can access management functions.
- The system blocks unauthorized actions.
- User permissions are enforced consistently.

User Story 31 - Inactive Session Automatic Lock Out

As the system, I need to end inactive sessions, so that an unattended system stays secure.

Acceptance Criteria

- Sessions expire after a specified period of no activity.
- Users must log in again after the session has timed out.
- Automatic lock out immediately ends system access.
- Users must log in again to resume their user session

Testing Techniques

Static Testing Techniques

Static testing techniques are specifically geared towards detecting defects early on in development by using methods which analyze the user requirements/stories, system designs and the actual source code without executing the code.

These techniques can find many issues before they become larger problems later in the development process and save the project time and money. In my test plan I will outline many static testing techniques which should be used and explain where and why they should be used.

Informal Reviews

Informal reviews are great at finding easy to spot mistakes by getting an additional pair of eyes on the issue. Buddy checks and pair reviews could spot a lot of the simpler issues and allow them to be resolved early on in testing, so a lot of time does not need to be spent resolving them later.

These types of techniques would be very effective at spotting issues spelling or grammar issues, unclear validation messaging, missing required indicators for required fields, and usability issues before dynamic testing takes place which would confirm at least some of these problems but would require more work to fix at that point.

User Stories which would benefit from informal reviews.

- **User Story 1 - View Current Benefit Selections**

This user story is about allowing the employee to view their current benefits coverage. A quick review by the tester and at least one other person could spot simple and obvious issues which would affect the employee's ability to view the benefits coverage summary and allow it to be resolved early on.

- **User Story 6 - Benefits Enrollment Form**

This user story is about allowing the employee to view the benefits enrollment form. Informal review could potentially find simple issues with this type of requirement.

- **User Story 11 - Display Cost Responsibilities**

This user story is about showing the employee their benefits costs versus the company-covered costs. If we know what should be covered by whom a quick review could spot mistakes.

- **User Story 12 - Prevent Form Submission When the Data is Incomplete**

This user story is about the employee being prevented from submitting their benefits enrollment form unless all the required information has been entered. If we know what the required information fields should be then a quick review could spot mistakes where a field that should be required is not or vice versa.

- **User Story 22 - Display Plan and Tier Cost Breakdowns**

This user story is about displaying a summary of the employees chosen benefits as a cost breakdown for each chosen plan and tier.

A review could potentially spot things which should be added or removed from the acceptance criteria.

- **User Story 24 - Validate the Employee's Data**

This user story is about the system validating that no employee information is missing.

This is something that a simple review could detect an issue with. A review could also determine if the logic for displaying an error message is sound.

- **User Story 25 - Validate Employee Dependent Relationships**

The user story is to validate that all the employee's dependents are being correctly added to the system and that no duplicates of the same dependent are being added by mistake.

An informal review could easily spot whether a relation to the employee exists, or an invalid entry should be rejected without needing to run the code to find out.

- **User Story 28 - Clear Error Messages**

This user story focuses on presenting the employee with clear easy to understand error messaging when there is a problem.

A review could determine if the error messages which will be delivered to the employee based on each error type will make perfect sense to everyone or if they are ambiguous and should be adjusted so they are easier to understand.

- **User Story 31 - Inactive Session Automatic Logout**

This user story is about the system automatically logging out of a session which is inactive for a predetermined amount of time.

A review of the steps taken by the system could reveal whether the requirement has valid logic even without being required to run any code or for that matter even without writing any code yet.

Walkthrough

A walkthrough is a bit more planned than just the standard pair testing and really is mostly informative where those that are most knowledgeable about certain areas of the project share their knowledge and explain the different aspects of the project in a structured meeting.

User Stories which would benefit most from a walkthrough.

- **User Story 2 - Modify Benefit Selections During the Open Enrollment**
- **User Story 3 - Add or Remove Dependents**
- **User Story 4 - Require Employee Coverage Before Dependent Coverage**
- **User Story 5 - Determine Coverage Tier Automatically**
- **User Story 13 - Remove My Coverage Removes My Dependents Coverage**
- **User Story 14 - Save Benefits Selections Draft**
- **User Story 15 - Confirmation and Submission of Benefits Selections**
- **User Story 16 - Searching and Viewing of Employee Records**
- **User Story 17 - Managing Dependents Records**
- **User Story 18 - Maintain Department Information**
- **User Story 19 - Inactivate Plan Options**
- **User Story 23 - Enforcing The Benefits Enrollment Period**
- **User Story 26 - Handling Coverage Tier Changes**

These stories are all process-driven and the opportunity for misunderstanding or of the user stories being interpreted by different people in different ways is higher than in most of the other user stories so having someone that is most familiar with each of the processes and how they will be used explain them in detail for everyone would be beneficial to the

team as a whole and give everyone a greater and more importantly unified understanding of the project, processes and the applications usage in real day to day use scenarios.

This will help to prevent building a system which meets all the technical requirements but does not meet the actual business needs of those that will be using it.

Technical Review

A technical review is structured review attended by both team members and technical experts used to evaluate the logic, calculations, integrations, and enforcement of business rules. Everything is well documented during this type of review.

These user stories are the main reasons why a technical review was chosen.

- **User Story 5 - Determine Coverage Tier Automatically**
- **User Story 7 - Update Effective Date**
- **User Story 10 - Maintaining Rates Tables**
- **User Story 20 - Calculate Benefits Costs**
- **User Story 21 - Apply the Effective Date Amounts**
- **User Story 22 - Display Plan and Tier Cost Breakdowns**
- **User Story 26 - Handling Coverage Tier Changes**
- **User Story 27 - Track Changes**
- **User Story 30 - Enforce Role-Based Access**

All these stories are primarily focused on performing calculations which if incorrect could cause serious issues, so they require more scrutiny than some of the other user stories to make certain of their correctness.

Some of these stories are also used for audit purposes which only place even more emphasis on the fact that these must be functioning correctly.

Inspection

An inspection is the most formal static review process and since some of the user stories for this application could have high-risk and compliance-sensitive functionality it just makes sense to treat the user requirements dealing with those areas of the application as carefully as we can to be sure we are not going to run into any costly legal issues.

- **User Story 4 - Require Employee Coverage Before Dependent Coverage**
- **User Story 7 - Update Effective Date**

- **User Story 20 - Calculate Benefits Costs**
- **User Story 21 - Apply the Effective Date Amounts**
- **User Story 23 - Enforcing The Benefits Enrollment Period**
- **User Story 27 - Track Changes**
- **User Story 29 - Authenticate Users**
- **User Story 30 - Enforce Role-Based Access**
- **User Story 31 - Inactive Session Automatic Logout**

All these user stories present highest risk, or they have some form of compliance-related factor attached to them, so it is important to treat them carefully even though the inspection process is a bit slower than the other forms of static testing.

Dynamic Testing Techniques

While static testing will find many errors early in development those types of testing can only test what an application should be doing based on requirements and it can only detect much, it is left to dynamic testing to really dig in and find out exactly how the application is actually functioning because what an application should be doing and what it is actually doing are not always the same.

If it was then there would be no need to run the code to test an application.

Black-box testing

This testing technique examines the functionality of application to determine if what the application is expected to do in a given situation is what it is actually doing performed from the point of view of the user, in all these cases as an employee, as a human resources employee or as the system itself to determine if it is behaving as it should.

All User stories have acceptance criteria which can be used as expected outcomes for generating use case black box testing so every user story will generate at least one multi-step pass or fail test scenario.

Equivalence Partitioning and Boundary Value Analysis

Equivalence partitioning and boundary value analysis can be used to determine what the application considers a valid or invalid user input such as numeric character where there shouldn't be any or other input format validation issues.

- **User Story 3 - Add or Remove Dependents**
- **User Story 7 - Update Effective Date**
- **User Story 8 - Maintain Employee Data**
- **User Story 9 - Maintain Insurance Company Data**
- **User Story 10 - Maintaining Rates Tables**
- **User Story 18 - Maintain Department Information**
- **User Story 24 - Validate the Employee's Data**
- **User Story 25 - Validate Employee Dependent Relationships**
- **User Story 29 - Authenticate Users**

All these user stories contain elements which require validation of some form or another which make them likely candidates for this type of testing.

Decision table testing

A decision table will be used to highlight exactly which combinations of circumstances are possible. It will illustrate in a table what should be an allowable combination of tiers, types, dependents, and coverage and what should not.

State transition testing

There are several state transitions within this application which must be tested. Whether open enrollment becomes locked once it ends, whether a successful login is transitioned to a valid session, or if an invalid login is rejected.

There are several other great reasons I am sure but the ones I have list are already reason enough to test these user stories listed below for state transition.

- **User Story 14 - Save Benefits Selections Draft**
- **User Story 15 - Confirmation and Submission of Benefits Selections**
- **User Story 23 - Enforcing The Benefits Enrollment Period**
- **User Story 29 - Authenticate Users**
- **User Story 31 - Inactive Session Automatic Logout**

All these user stories deal with conditional transitions of the applications state.

Role-based access testing

The application will have more than one role type and at least one of the roles defined in the scenario for this application has an elevated level of permissions which should not be available to the average employee because of this it makes good sense from a system security stand point to test whether user roles are being correctly applied to a standard employee user role versus an human resources employee user role when they are logged in to the system.

Integration testing

This application will have many components which all must function as a single application together.

Almost all parts of this application must have interconnectivity with other parts of the application, databases, or documents which must be retrieved or stored.

There are many elements which must be tested for integration and communication functionality to be certain they are working correctly together.

Without proper integration testing the wrong files or incorrect data could be retrieved, the application could fail to connect to the database storing the employee's information preventing them from viewing their current benefits coverage or making changes, or maybe the HR employees could potentially discover that they cannot retrieve the necessary information about employees required to do their job.

Regression testing

Regression testing is invaluable during development. We spend a lot of time and effort making certain that things are working correctly as we are coding them when making an application such as this and as things are added new changes can break things which are already working correctly.

Regression testing is necessary to help prevent these unforeseen collateral damage situations when changes are made and new things are working great but something else gets broken at the same time from happening without being noticed right away. Ideally

regression tests will be automated and run after each new change to be certain that everything is still functional.

Exploratory testing

Structured testing cannot always find the outside the box issues. This is where exploratory testing shines because its purpose is to think outside the box and find the issues that would never happen under normal usage circumstances.

Finding edge cases is the ultimate goal of exploratory testing and while it may not seem like a priority, the edge case errors are often the most damaging in an application so especially in an application such as this, which deals with personal information, benefits and payroll it must be performed thoroughly to find as many edge case issues as possible so they can be fixed before they are a serious problem.

Test Conditions, Test Cases, and Potential Test Scripts

Test Condition 1

Viewing Current Benefit Elections

This test condition is to verify that an employee can view their current benefits, their dependents, and that the system is clearly communicating when no benefits exist.

User Stories

- User Story 1 - View Current Benefit Selections

Acceptance Criteria

- The application displays all selected benefit plans for the employee.
- All covered dependents are listed under each applicable plan.
- The plan type of PPO or HMO is clearly identified.
- If the employee has no benefit selections, the system clearly displays a “no benefits are selected” message.

Priority Level 1 - High

This condition is assigned as a priority of high because while it does not modify the data, it is a main function for the usability and validation of employee benefit selections

and usage of the application. If employees cannot accurately view their benefits, they cannot verify they are correct before submission, which could lead to benefit enrollment errors and cause HR support issues.

Test Condition 2

Modifying Benefits During the Enrollment Period

This test condition confirms that employees are able to modify benefit selections only during the allowed enrollment window and that all changes are saved and reflected correctly.

User Stories

- User Story 2 - Modify Benefit Selections During the Open Enrollment
- User Story 23 - Enforcing the Benefits Enrollment Period

Acceptance Criteria

- The system allows benefit plan changes only during the enrollment period.
(User Story 2)
- The employee can switch between PPO and HMO for health care coverage.
(User Story 2)
- Optional dental and vision coverage can be added or removed.
(User Story 2)
- Changes are saved immediately in the benefits summary which is displayed.
(User Story 2)
- Benefit selections can only be altered during the allowed enrollment period.
(User Story 23)
- Outside of the enrollment period, the benefits information is read-only.
(User Story 23)

Priority Level 0 - Critical

This is a critical priority because the open benefits enrollment functionality is the main purpose of the application. If employees cannot modify benefits correctly or are allowed to change them outside of the enrollment window, the system would be violating the company's benefits enrollment policy and potentially create a legal and payroll issue.

Test Condition 3

Managing Dependents Within Benefit Coverage

This test condition verifies that dependents can be added or removed correctly, required data is validated, and duplicate or invalid dependent records are prevented from being added.

User Stories

- User Story 3 - Add or Remove Dependents
- User Story 25 - Validate Employee Dependent Relationships

Acceptance Criteria

- The system allows entry of new dependent names, their relationship, and all required personal information.
(User Story 3)
- Dependents can be removed from the benefits coverage selections.
(User Story 3)
- The system prevents duplicate dependent entries.
(User Story 3)
- A relationship type must be selected for all dependents.
(User Story 25)
- Duplicates of a dependent are not allowed.
(User Story 25)

Priority Level 0 - Critical

Dependent handling directly impacts eligibility and cost calculations. Errors in this area could result in invalid coverage, financial discrepancies, or compliance violations, which makes this a critical priority function.

Test Condition 4

Enforcing Coverage Eligibility Rules

This test condition verifies that dependents cannot be enrolled in coverage unless the employee is also enrolled in the same coverage and that removing employee coverage removes all dependent coverage automatically.

User Stories

- User Story 4 - Require Employee Coverage Before Dependent Coverage
- User Story 13 - Remove My Coverage Removes My Dependents Coverage

Acceptance Criteria

- The system prevents selecting dependent coverage if employee benefits coverage is not yet selected.
(User Story 4)
- Invalid selections cannot be saved.
(User Story 4)
- Removing an employee's coverage automatically removes their dependents coverage.
(User Story 13)
- The system prevents any orphaned dependents benefits selections.
(User Story 13)

Priority Level 0 - Critical

This is a critical priority test case because if violated, the system could allow an illegal or invalid enrollment of insurance, creating serious compliance, legal and or financial risks for the company.

Test Condition 5

Automatic Coverage Tier Determination

This test condition validates that the system assigns and recalculates the correct coverage tier when dependents are added or removed.

User Stories

- User Story 5 - Determine Coverage Tier Automatically
- User Story 26 - Handling Coverage Tier Changes

Acceptance Criteria

- The system assigns the correct coverage tier based on the selected dependents.
(User Story 5)
- The coverage tier is updated automatically when dependents are added or removed.
(User Story 5)
- The correct coverage tier is always displayed based on the dependents included.
(User Story 26)

- The benefits costs are adjusted accordingly.
(User Story 26)

Priority Level 0 - Critical

Benefits coverage tiers, number of dependents, and plan types are all things which would affect the cost calculations so any unresolved issues in these areas will result in incorrect billing and payroll deductions, this warrants a critical priority level because of that.

Test Condition 6

Benefit Cost Calculation and Rate Application

This test condition makes certain that costs are being calculated using the correct rate tables, that effective dates are being applied and that the benefits cost totals are being updated whenever a selection is changed.

User Stories

- User Story 20 - Calculate Benefits Costs
- User Story 21 - Apply the Effective Date Amounts
- User Story 22 - Display Plan and Tier Cost Breakdowns

Acceptance Criteria

- Benefits costs are calculated using the correct rate table.
(User Story 20)
- Cost calculations are updated when the selections have changed.
(User Story 20)
- The system selects proper rates based on the enrollment's effective date.
(User Story 21)
- A total monthly cost is shown.
(User Story 22)

Priority Level 0 - Critical

Incorrect cost calculation affects payroll, accounting, and the employee's trust in the application. Financial accuracy is of critical importance in this application so this earns a critical priority.

Test Condition 7

Enrollment Form Generation and Submission Validation

This test condition verifies that enrollment forms are submitted accurately and that they cannot be submitted unless all of the required information has been completed.

User Stories

- User Story 6 - Benefits Enrollment Form
- User Story 12 - Prevent Form Submission When the Data is Incomplete
- User Story 15 - Confirmation and Submission of Benefits Selections

Acceptance Criteria

- The system generates a form reflecting the current benefits selections.
(User Story 6)
- The form cannot be submitted if required information is missing.
(User Story 6) (User Story 12)
- The employee must approve the selections before submission occurs.
(User Story 15)
- Submitted benefits selections are prevented from any further editing.
(User Story 15)

Priority Level 1 - High

It is not as critical as the calculations test conditions are , this test condition is a high priority because it is making certain there is data integrity before the benefits enrollment is allowed to be finalized.

Test Condition 8

Authentication, Authorization, and Session Security

This test condition verifies that a secure authenticated login is taking place before a user session can begin, role-based access is being enforced, and that the automatic session timeout behavior is working correctly.

User Stories

- User Story 29 - Authenticate Users
- User Story 30 - Enforce Role-Based Access
- User Story 31 - Inactive Session Automatic Logout

Acceptance Criteria

- The user must provide valid credentials.
(User Story 29)
- Regular employees cannot access administrative functions.
(User Story 30)
- Sessions expire after a specified period of no activity.
(User Story 31)

Priority Level 0 - Critical

Any security failures could expose sensitive personal information such as benefits data and create legal issues and privacy risks. Therefore, this test condition must be considered a critical priority.

Test Condition 9

PPO vs HMO Plan Type Identification

This test condition is designed to verify that when an employee views their medical coverage, the system is clearly identifying whether the medical plan which is selected is a PPO or an HMO.

User Stories

- User Story 1 - View Current Benefit Elections
- User Story 2 - Modify Benefit Elections During the Open Enrollment

Acceptance Criteria

- The plan type of PPO or HMO is clearly identified.
(User Story 1)
- The employee can switch between PPO and HMO for health care coverage.
(User Story 2)

Priority Level 2 - Medium

This test condition is important for the decision making process of the employee but its functions are informative not calculative so this test condition only warrants a medium priority level.

Test Condition 10

Dependent Adding or Removing Updates Benefits Summary Immediately

This test condition is designed to verify that when a dependent is added or removed, the benefits summary will be updated immediately and reflects the correct dependent coverage information.

User Stories

- User Story 3 - Add or Remove Dependents
- User Story 1 - View Current Benefit Elections
- User Story 2 - Modify Benefit Selections During the Open Enrollment

Acceptance Criteria

- Updates immediately adjust the benefits coverage summary.
(User Story 3)
- All covered dependents are listed under each applicable plan.
(User Story 1)
- Changes are saved immediately in the benefits summary which is displayed.
(User Story 2)

Priority Level 1 - High

This condition is high priority because the benefits summary is the main method employees are using to verify their benefit selections before they submit their benefits selection. An incorrect or delayed update here could cause the employee to submit incorrect benefit selections.

Test Condition 11

Updated HR Effective Date Are Shown and Are Logged

This test condition confirms that HR can update effective dates for benefits, that the date appears correctly on the benefits enrollment form, and that the change is recorded so there is a stored history in case of audit.

User Stories

- User Story 7 - Update Effective Date
- User Story 27 - Track Changes

Acceptance Criteria

- Only a Human Resources employee can edit the effective date.
(User Story 7)
- The updated date appears on the enrollment form.
(User Story 7)
- All changes are recorded in the system.
(User Story 7)
- The system records who made each benefits change.
(User Story 27)
- The date and time of the changes are stored.
(User Story 27)

Priority Level 0 - Critical

Effective dates and audit logging can impact coverage start dates, payroll timing, and compliance which means that coverage may not begin when it should and there is inaccurate traceability for auditing purposes. These reasons make this a critical priority.

Test Condition 12

Employee Record Maintenance and Validation

This test condition confirms that an HR employee can maintain an employee's records, that required fields are being enforced, that formats are validated, and that duplicate employee IDs are being prevented from being created.

User Stories

- User Story 8 - Maintain Employee Data
- User Story 24 - Validate the Employee's Data

Acceptance Criteria

- Required fields include employee ID, department ID, name, address information, and office extension number.
(User Story 8)
- The system validates required fields.
(User Story 8)
- Duplicate employee IDs are not allowed.
(User Story 8)
- Required data fields must be completed.
(User Story 24)

- Data field formats are being validated.
(User Story 24)
- Duplicate employee records are being prevented.
(User Story 24)
- Any errors are being clearly communicated.
(User Story 24)

Priority Level 1 - High

This test condition is a high priority because while these functions not working correctly may not prevent usage they do make sure that the employee information is accurate and allow HR employees to correct any information which is entered in error.

Test Condition 13

Maintenance of Insurance Providers Including Their Inactivation

This test condition verifies that an HR employee can add or edit insurance providers, that the system prevents duplicate company numbers, and that providers can be made inactive without deleting their data history.

User Stories

- User Story 9 - Maintain Insurance Company Data

Acceptance Criteria

- Insurance company numbers and company names can be added or edited.
- The system prevents duplicating company numbers.
- Existing enrollment records are not affected by updates.
- Insurance providers can be marked as inactive without being deleted.

Priority Level 2 - Medium

This test condition is focused on administration functions more than employee enrollment useability, so it only has a priority level of medium.

Test Condition 14

Rate Table Maintenance by Plan, Tier, and Benefits Effective Period

This test condition confirms that an HR employee can define benefit rates by plan and or tier. that invalid values are being rejected, that the rates only apply to the correct benefits period, and that changes are affecting the calculations as they are intended.

User Stories

- User Story 10 - Maintaining Rates Tables
- User Story 21 - Apply the Effective Date Amounts

Acceptance Criteria

- Rates can be defined for each plan and tier combination.
(User Story 10)
- Invalid rate values are rejected.
(User Story 10)
- Updates apply only to the effected period.
(User Story 10)
- Benefits rates are associated with the defined period.
(User Story 21)
- Any expired benefit rates are not being used.
(User Story 21)
- Future rates are applied when they take effect.
(User Story 21)

Priority Level 0 - Critical

Rates are what determines if the cost of benefits is accurate. If the rate and period rules are incorrect the employee can be charged incorrectly. Which makes this a critical priority.

Test Condition 15

Plan and Tier Combination Validation and Rejection

This test condition is to make certain that any invalid plan and tier combinations are not allowed by the system and they are rejected before they affect benefits total or submission.

User Stories

- User Story 20 - Calculate Benefits Costs

Acceptance Criteria

- Any invalid plan and tier combinations are rejected.
- Benefits cost totals match defined plan and tier rates.

Priority Level 0 - Critical

Allowing an invalid plan and tier combination would produce an incorrect total and potentially create invalid enrollments so this is a critical priority level test condition.

Test Condition 16

Company-Paid vs Employee-Paid Cost Responsibility Display

This test condition verifies the application is clearly distinguishing between which benefits are company-paid and which benefits are employee-paid such as dental and vision so that accurate benefits cost totals are being displayed to the employee.

User Stories

- User Story 11 - Display Cost Responsibilities
- User Story 22 - Display Plan and Tier Cost Breakdowns

Acceptance Criteria

- Medical costs which apply are displayed as company paid.
(User Story 11)
- Dental and vision benefits costs are displayed as employee paid.
(User Story 11)
- A clear benefits cost breakdown is shown.
(User Story 11)
- Totals are updated when benefits selections have been changed.
(User Story 11)
- The Plan and Tier costs are displayed by coverage type.
(User Story 22)
- A total monthly cost is shown.
(User Story 22)

Priority Level 1 - High

Employees must fully understand their benefits costs in order to make an informed decision about which plans they want to select. The ability to receive accurate totals is important but will not prevent the use of this application so this receives a priority rating of high.

Test Condition 17

Saving and Restoring Incomplete Benefits Elections

This test condition confirms that incomplete benefits selections can be saved as a draft and then restored later when the employee wants to resume their benefits elections, and that this will not affect any active benefits until the benefits selections draft has been actually submitted.

User Stories

- User Story 14 - Save Benefits Selections Draft

Acceptance Criteria

- The system allows the saving of incomplete benefits selections as a draft.
- Draft data is restored when the employee returns.
- Drafts do not affect active benefits enrollment until they have been submitted.
- Employees can continue editing a saved draft.

Priority Level 2 - Medium

Draft saving is a quality-of-life feature and reduces abandonment during enrollment. It's important, but not as critical as rule enforcement, calculations, or security.

Test Condition 18

Confirmation, Submission, and Post-Submission Edit Prevention

This test condition verifies that the confirmation screen is being shown and that benefits selection submission requires an approval, also that benefits submissions are being recorded, that the users are notified, and lastly that benefits can no longer be edited after the submission.

User Stories

- User Story 15 - Confirmation and Submission of Benefits Selections

Acceptance Criteria

- A confirmation screen summarizes all benefit selections.

- The employee must approve the benefits selections before submission occurs.
- Submitted benefits selections are prevented from any further editing.
- The submission of the selected benefits is recorded successfully.
- The employee is notified on the display that the submission was successfully recorded.

Priority Level 1 - High

The point of the benefits submission is the last moment when changes can be made before enrollment. Because of this a priority of high is assigned to this test condition.

Test Condition 19

Department Maintenance and Assignment Rules

This test condition confirms that an HR employee can add and update departments, that an invalid department cannot be assigned to an employee, and departments which are currently in use cannot be deleted.

User Stories

- User Story 18 - Maintain Department Information

Acceptance Criteria

- Departments can be added or updated.
- Invalid departments cannot be assigned to employees.
- Departments which are in use cannot be deleted.
- Updates are saved immediately.

Priority Level 3 - Low

Departments are not as critical as the benefits enrollment rules and their costs or login and authentication, so this test condition receives a low priority level.

Test Condition 20

Audit Logs Are Complete and Immutable

This test confirms that audit logs capture the who, what and when of all changes made so there is a clear audit trail to follow should they be needed and that audit data cannot be edited.

User Stories

- User Story 27 - Track Changes

Acceptance Criteria

- The system records who made each benefits change.
- The date and time of the changes are stored.
- Any modified fields are identifiable.
- Auditable data is read only.

Priority Level 1 - High

Auditability is incredibly important for human resources accountability and for compliance.

While the lack of a clear audit trail does not prevent the application from being used it still warrants assigning this test case a priority of high because it has potential legal and compliance ramifications if audit data is needed and cannot be produced.

Test Condition 21

Role-Based Function Restriction and Unauthorized Action Blocking

This test condition is to confirm that the system is enforcing all role based permissions consistently, which is to prevent regular employees from accessing and perform administrative actions and allows HR employees to access and use management functions.

User Stories

- User Story 30 - Enforce Role-Based Access

Acceptance Criteria

- Regular employees cannot access administrative functions.
- Human Resources employees can access management functions.
- The system blocks unauthorized actions.
- User permissions are enforced consistently.

Priority Level 0 - Critical

Role enforcement protects sensitive data and prevents an unauthorized user from performing modifications to the benefits and rates of their own or other employee's records. This is a security related test condition and because of that it earns a critical priority.

Test Condition 22

Authentication and Invalid Login Handling

This test condition confirms that a valid set of credentials are required to access the application user session, that an invalid login is rejected, and a user session can only begin after a successful user credentials authentication.

User Stories

- User Story 29 - Authenticate Users

Acceptance Criteria

- The user must provide valid credentials.
- All invalid login attempts are rejected.
- Access to the system is only granted after authentication.
- A session only begins after login is successful.

Priority Level 0 - Critical

Authentication protects all employees and their dependent information. If authentication were to fail, the application would become insecure, and anyone may be able to access personal data about others. For this reason, this is a critical priority test condition.

Test Condition 23

Timeout of Inactive Sessions and That Re-Authentication is Forced

This test condition verifies that any inactive session will be expired after a specified amount of inactivity and the user will automatically be locked out and forced to log in again before they can resume their user session.

User Stories

- User Story 31 - Inactive Session Automatic Lock Out

Acceptance Criteria

- Sessions expire after a specified period of no activity.
- Users must log in again after the session has timed out.
- Automatic lock out immediately ends system access.
- Users must log in again to resume their user session

Priority Level 1 - High

User session timeouts prevent unauthorized access to the system if a user session is left logged in but is unattended. This is a high priority level test condition.

Test Cases

Assumptions For All Test Cases

- The benefits application is a web app.
- All tests are performed using Chrome web browser.
- There are test credentials for both a standard employee and an HR employee which are.
login: Emp1 pw: password1 for the standard employee.
And
Login: HR1 pw: password2 for the HR employee.
- The tester has opened the web app to the correct URL and the login screen is loaded.

Test Case 1 - View Current Benefit Elections

Priority Level 1 - High

Employees must be able to view benefit selections to verify correctness before submission. If this fails, enrollment errors and HR support issues increase even if the app still loads.

My Assumptions Made for This Test Case

- The employee is already enrolled in at least one plan.
- The employee has at least one dependent that is covered under at least one plan.
- The benefits summary page is accessible after employee login.

Purpose

Verify the benefits summary displays the enrolled plans, covered dependents under each enrolled plan, and clearly identifies if the plan type is a PPO or an HMO.

Test Steps

1. Login as a standard employee.
2. Navigate to “Benefits Summary and Current Elections”.
3. Record the medical plan type shown. Record the result (A).
4. Verify dependents displayed under each applicable plan. Record the result (B).

Expected Results

- A) PPO or HMO is clearly displayed for the medical coverage type.
- B) All covered dependents appear under each plan where they are enrolled.

Why This Test Was Chosen

This is the main employee verification screen.

Test Case 2 - Modify Benefits During Enrollment Period

Priority Level 0 - Critical

Open enrollment changes are the main purpose of the application. If changes cannot be made during enrollment or if saving fails, the app fails its primary objective.

My Assumptions Made for This Test Case

- The system’s current date and time is within the defined enrollment period.
- The employee has an existing medical plan selection.
- Save is triggered by a button and data persists once it is pressed.

Purpose

This test case verifies that an employee can change their benefits during the enrollment time period and that changes are saved and reflected immediately.

Test Steps

- 1) Login as a standard employee.
- 2) Navigate to benefits elections edit screen.
- 3) Switch medical benefit options from PPO to HMO.
- 4) Add dental coverage.

- 5) Remove the vision coverage.
- 6) Navigate to summary screen. Record the result (A).
- 7) Logout of the application and then login again and return to benefits summary.
Record the result (B).

Expected Results

- A) The summary immediately shows that an HMO and dental benefits selection has been changed and has updated the benefits total cost.
- B) After login again, changes made from before the logout remain saved and are accurate.

Why This Test Was Chosen

This verifies that an employee can make benefits changes and that those changes will persist even in a new user session.

Test Case 3 - Add a Dependent with the Required Fields and Prevent Duplicates

Priority Level 0 - Critical

Proper dependent handling will affect both eligibility and the benefits cost calculations. A duplicate dependent or missing required information can create an invalid coverage entry.

My Assumptions Made for This Test Case

- A duplicate dependent is the same individual being listed more than once and the systems validations will use unique identifiers to prevent duplicate dependents from being added.
- A relationship is required for all dependents.
- An employee is allowed to add their dependents during enrollment.

Purpose

To verify that adding a dependent is allowed, that validation prevent leaving required fields blank and that the system is preventing duplicate dependents from being added.

Test Steps

- 1) Login as a standard employee.
- 2) Navigate to dependent management.
- 3) Add a new dependent with all the required fields completed.

- 4) Save. Record the result (A).
- 5) Attempt to add the same dependent using the same information again.
- 6) Save. Record the result (B).

Expected Results

- A) The dependent is saved and is visible in the employee record.
- B) The duplicate dependent is blocked from being saved and a clear message shown that a duplicate dependent is not allowed and the duplicate record is not created.

Why This Test Was Chosen

It covers both the valid addition of a dependent and the prevention of a duplicate dependent entry as well as verifying that required fields validation is working properly.

Test Case 4 - Dependent Coverage Blocked Until Employee Coverage Exists and Orphan Dependent Prevention

Priority Level 0 - Critical

If this rule fails, the system could create invalid or illegal coverage enrollments and orphaned dependent selections.

My Assumptions Made for This Test Case

- An employee must select the same coverage type before selecting dependents to be added to that coverage.
- The system prevents saving any invalid coverage combinations.

Purpose

This verifies that dependents cannot be enrolled in a benefit plan unless the employee is first enrolled in the same plan, and that invalid benefits selections cannot be saved.

Test Steps

- 1) Login as a standard employee.
- 2) Make sure the employee does not have any dental coverage added.
- 3) Attempt to enroll a dependent in a dental plan without enrolling employee in a dental plan.
- 4) Observe the application's behavior. Record the result (A).
- 5) Attempt to save the changes. Record the result (B).

Expected Results

- A) The system blocks the dependent dental plan selection and shows a clear validation message.
- B) The invalid selection cannot be saved, and no employee or dependent dental coverage is listed in benefits summary.

Why This Test Was Chosen

This test case validates the most important rule with regard to eligibility and prevents invalid records and coverage from occurring.

Test Case 5 - Employee Coverage Tier Determination and Automatic Benefits Cost Recalculation

Priority Level 0 - Critical

The coverage tier is a main factor in determining benefits cost. If the automatic selection of tiers based on the selected dependents has a logic failure, the benefits cost summary and the employees payroll deductions can be wrong.

My Assumptions Made for This Test Case

- The tier definitions are (employee only), (employee and spouse), (employee, spouse and one child), (employee and family).
- The family tier is applied only when the employees' number of dependents is more than a spouse and one child.

Purpose

This verifies that each tier is being correctly assigned and that the tier selection updates automatically when the number of dependents changes.

Test Steps

1. Login as a standard employee.
2. Select a PPO medical.
3. Add a spouse and one child to the medical coverage.
4. Observe the tier shown. Record the result (A).
5. Add a second child to the medical coverage.

6. Observe the tier shown. Record the result (B).

Expected Results

- A) The tier selected is (employee, spouse and one child).
- B) The tier has been recalculated automatically to select (employee and family).

Why This Test Was Chosen

This test case verifies the tier transitions are functioning, and it confirms that the tier updates are occurring automatic based on the dependents added.

Test Case 6 - Enrollment Form Creation and Block Submission When Incomplete

Priority Level 1 - High

Enrollment submission must be blocked when required data is missing in order to maintain data integrity.

My Assumptions Made for This Test Case

- The enrollment form pulls its data from the employee's current new benefits selections data such as employee information, listed dependents and their benefits selections and then compiles the data pulled into a benefits summary.
- The system highlights any missing data for required fields and shows messages detailing what information is still missing.

Purpose

This confirms that the summary form reflects all current benefit selections and that it cannot be submitted with any missing required information.

Test Steps

- 1) Login as a standard employee.
- 2) Navigate to enrollment form summary and verify it includes employee information, all dependents and the correct benefits plan selections. Record the result (A).
- 3) Remove the entry in the required employee address line.
- 4) Attempt to submit form. Record the result (B).

Expected Results

- A) The summary form accurately reflects the current selections and dependent info.

- B) The submission is blocked and any missing fields are highlighted, and a message explains what corrections needed before the form can be submitted.

Why This Test Was Chosen

This test case confirms that the benefits summary form cannot be submitted with missing information that is required for maintaining accurate records and correctly assigning benefits coverage to the employee and their dependents.

Test Case 7 – Authentication, Role-Based Access and Session Timeout

Priority Level 0 - Critical

If authentication or authorization fails, sensitive HR only employee benefits and personal data will be exposed.

My Assumptions Made for This Test Case

- There are at least two user roles which are employee and HR.
- There is a user session inactivity timeout value which is after 5 minutes of inactivity.

Purpose

The purpose is to verify that a valid login is required to access the application, confirm that HR employee functions are blocked for standard employees, and that user sessions will expire after 5 minutes of inactivity.

Test Steps

- 1) Attempt access to app by entering invalid user credentials. Record the result (A).
- 2) Login as a standard employee and attempt to access the HR admin functions. Record the result (B).
- 3) Login as an HR employee and try to access HR administrative functions. Record the result (C).
- 4) While still logged in remain inactive until the 5-minute timeout threshold has passed. Record the result (D).

Expected Results

- A) Access is prevented and you are presented with a message that your login or password is invalid.
- B) The standard employee is blocked from accessing administrative functions.
- C) The HR employee can access HR administrative functions.

- D) The user session expires and the user is locked out they must login again to continue their user session.

Why This Test Was Chosen

This test case confirms that the primary method of securing this application is functioning correctly.

Test Case 8 - PPO versus HMO Identification on the Benefits Summary After Switching

Priority Level 2 - Medium

The plan identification is key for allowing the employee to make an informed decision regarding their benefits, if the incorrect information is displayed to them they may make mistakes in their selections during their benefit enrollment.

My Assumptions Made for This Test Case

- The medical plan types are clearly labeled either PPO or HMO when they appear on the benefits summary and confirmation screens.
- When switching plan types the summary will update immediately to reflect the change.
- The employee has current medical coverage and it is a PPO.

Purpose

To confirm that the correct information about PPO and HMO current selections is being displayed to the employee.

Test Steps

- 1) Login as a standard employee.
- 2) Verify current medical plan type is displayed. Record the result (A).
- 3) Switch the plan type from PPO to HMO.
- 4) Return to benefits summary and observe the displayed plan type. Record the result (B).

Expected Results

- A) The medical plan type is clearly labeled as a PPO.
- B) After switching the plan type the benefits summary shows the new plan type HMO.

Why This Test Was Chosen

The purpose of this application is to provide the employees with accurate information so they can make informed decisions and select their benefits coverage. If the information they are given is inaccurate they will base their decisions on false information.

Test Case 9 - Benefits Summary Updates Immediately After Dependent Changes

Priority Level 1 - High

Employees must rely on the benefits summary to confirm their selections before submission. Delayed or incorrect summary data can cause incorrect enrollment.

My Assumptions Made for This Test Case

- The benefits summary updates immediately.
- The employee is already enrolled in a dental plan for themselves.

Purpose

To confirm that the benefits summary updates immediately when dependents are added or removed.

Test Steps

- 1) Login as a standard employee.
- 2) Add a dependent under dental coverage.
- 3) Navigate to the benefits summary and confirm that the dependent appears. Record the result (A).
- 4) Remove the dependent from dental coverage.
- 5) Check the benefits summary again to confirm they have been removed. Record the result (B).

Expected Results

- A) The added dependent appears under dental plan on benefits summary immediately.
- B) The removed dependent no longer appears on the benefits summary dental plan.

Why This Test Was Chosen

This test confirms both the accuracy and responsiveness of the summary verification mechanism for adding and removing dependents.

Test Case 10 - An HR Employee Can Change Effective Date And Updates Are Displayed and Logged

Priority Level 0 - Critical

The effective date controls when coverage starts.

My Assumptions Made for This Test Case

- Effective date is visible on the benefits enrollment summary.
- Audit logs exist and are viewable by HR.

Purpose

To verify that an HR employee can update the effective dates, the change appears on the benefits summary, and that an audit record is created for the change.

Test Steps

- 1) Login as an HR employee.
- 2) Search for Employee "EMP1" record.
- 3) Update effective date to "06/01/2026".
- 4) View employee enrollment summary and record effective date. Record the result (A).
- 5) View the audit log and confirm that a record for the entry exists. Record the result (B).

Expected Results

- A) The enrollment summary shows "06/01/2026" effective date.
- B) Audit log includes who changed it, date/time, and field changed.

Why This Test Was Chosen

This test case verifies that an HR employee is able to alter the effective date of an employee's benefits and that there is an auditable record of the change.

Test Case 11 - Add Employee Record and Validating Required Fields

Priority Level 1 - High

Accurate employee data is required for nearly every function in this application to work correctly. It must also be uniquely identifiable in order to be accessible among the other employees.

My Assumptions Made for This Test Case

- The employee ID must be unique.
- Required fields are enforced and error messages are shown clearly.

Purpose

The purpose of this test case is to confirm that an HR employee can update an employee's information, that any required fields are being validated by the system, and that assigning duplicate employee IDs are being prevented.

Test Steps

- 1) Login as an HR employee.
- 2) Edit employee EMP1's record and leave the address blank. Attempt to save. Record the result (A).
- 3) Enter valid address and save. Record the result (B).
- 4) Attempt to create a new employee using an existing employee ID EMP1. Record the result (C).

Expected Results

- A) The save is prevented and a required field error shown.
- B) The save is successful and the updated employee information is visible immediately.
- C) The duplicate employee ID is rejected with clear message stating that the employee ID must be unique.

Why This Test Was Chosen

This test verifies that an HR employee can edit an employee's information when they need to or even add a new employee, but that they must follow the requirements in place by the system such as not using an employee ID which is already in use and not leaving required fields blank.

Test Case 12 - Rate Table Maintenance by Benefits Period and Rejection of Invalid Values

Priority Level 0 - Critical

The benefit rates are the foundation of accurate costs within this application.

My Assumptions Made for This Test Case

- All rates must be a non-negative numeric value.
- All rates are specifically applied to specified benefits time date periods and will not overwrite another period.
- That all benefit periods are 1 calendar year from January 1st to December 31st
- The rates are per benefit and are charged per paycheck for the duration of benefits period.

Purpose

To confirm that an HR employee can enter and edit valid benefit rates, that invalid benefit rates are being rejected, and that benefit rates apply only to the selected benefits period.

Test Steps

- 1) Login as an HR employee.
- 2) Add a rate of (\$100.00) for medical PPO during the (2026) benefits period.
- 3) Save the entry. Record the result (A).
- 4) Edit the same rate and set it as (-\$15.00).
- 5) Save the entry. Record the result (B).
- 6) Add a rate of (\$110.00) for medical PPO during the (2027) benefits period.
- 7) Save the entry. Record the result (C).

Expected Results

- A) The valid rate is saved.
- B) The invalid negative rate is rejected with a message explaining that the rate cannot be a negative number.
- C) The 2027 rate is saved without affecting 2026 rate.

Why This Test Was Chosen

This test case verifies that an HR employee can edit or add a benefit but that only valid rates can be assigned for a benefit period.

Test Case 13 - Company-Paid vs Employee-Paid Cost Responsibility Display

Priority Level 1 - High

Employees must understand their payroll impact for their benefits. Incorrectly labeling cost responsibilities could cause disputes about an employee's benefits payroll deductions.

My Assumptions Made for This Test Case

- Medical for the (employee only) is shown as company-paid.
- Dental and vision are shown as employee-paid.
- Totals are updated as the benefits selections are changed.

Purpose

To confirm correct the labeling and breakdown of company vs employee costs.

Test Steps

- 1) Login as a standard employee.
- 2) Select medical, dental and vision.
- 3) View the benefits summary and observe the paid by labels and cost total. Record the result (A).
- 4) Remove vision and again view the benefits summary and observe the paid by labels and cost total. Record the result (B).

Expected Results

- A) Medical is labeled company-paid on the benefits summary while dental and vision are labeled employee-paid. The totals of the benefits displayed reflect the correct costs.
- B) The benefits total updates immediately reflect the change in benefits coverage selected. The totals of the benefits displayed reflect the correct costs.

Why This Test Was Chosen

This test case was chosen to test if the costs of benefits are employee paid versus company paid is being effectively communicated to the employee.

Test Case 14 - Department Maintenance and Assignment Rules

Priority Level 3 - Low

Department maintenance allows an HR employee to edit, add and remove departments in order to maintain data integrity but it is less critical than enrollment, cost, and security.

My Assumptions Made for This Test Case

- Departments in use cannot be deleted.
- An invalid department cannot be assigned.

Purpose

To confirm that an HR employee can add, update or delete a department when it is required and that the rules regarding the deletion or editing of departments is being enforced.

Test Steps

- 1) Login as an HR employee.
- 2) Add a department (Marketing). Save and record the result (A).
- 3) Attempt to assign an employee to a department which does not exist. Record the result (B).
- 4) Attempt to delete a department which is currently assigned to an employee. Record the result (C).

Expected Results

- A) The new department saved successfully.
- B) The invalid department assignment prevented.
- C) The deletion is blocked for the department because it is in use.

Why This Test Was Chosen

This validates basic administrator data integrity rules for departments.

Test Case 15 - Audit Logs Are Complete and Read-Only

Priority Level 1 - High

An auditable trail of actions supports compliance and dispute resolutions.

My Assumptions Made for This Test Case

- An audit log captures the who, what, and when of every change which is made to an employee's benefits.
- The audit logs cannot be edited.

Purpose

To confirm that audit logs are recording all events as well as who performed them and that they are immutable.

Test Steps

- 1) Login as an HR employee.

- 2) Make a change to the first benefit rate.
- 3) Open the audit log for the affected record. Record the result (A).
- 4) Attempt to edit or delete audit entry. Record the result (B).

Expected Results

- A) The audit log shows the user, the date, the time, and which fields were changed.
- B) The audit log entry cannot be edited or deleted and the read-only designation is enforced.

Why This Test Was Chosen

It makes certain there is an auditable trail of all changes and that tampering with audit logs is prevented.

Test Case 16 - Role-Based Restrictions Block Unauthorized Actions

Priority Level 0 - Critical

If employees can access HR management functions, sensitive data and system integrity are placed at risk.

My Assumptions Made for This Test Case

- The standard employee and HR employee roles already exist, and role-based permissions are being enforced consistently.

Purpose

To verify that standard employees cannot access administrative functions and the HR employees can.

Test Steps

- 1) Login as a standard employee. Attempt access to rate table maintenance and employee search. Record the result (A).
- 2) Login as an HR employee. Access those functions successfully. Record the result (B).

Expected Results

- A) The standard employee is blocked from admin functions, and a message is shown which explains why access was denied.
- B) The HR employee has access and can perform allowed administrative actions.

Why This Test Was Chosen

It validates role-based authorization control systems.

Test Case 17 - Authentication Rejects Invalid Login and Prevents Session from Starting

Priority Level 0 - Critical

User authentication protects all employee benefit and dependents personal data.

My Assumptions Made for This Test Case

- Invalid credentials will never start a session.
- An error message is shown if invalid credentials are used and no access is given to sensitive information.

Purpose

Verify that an invalid login is rejected and that a session can only begin after successful authentication.

Test Steps

- 1) Attempt login with valid username and wrong password (EMP1 and pw password2). Record the result (A).
- 2) Attempt to login with valid credentials (HR1 and pw password2). Record the result (B).

Expected Results

- A) The login is rejected and the user session not created.
- B) The login succeeds and the HR user session begins. The HR user can then access the areas they are authorized to view and change.

Why This Test Was Chosen

It verifies the login process which is protecting the system is preventing unauthorized access.

White-Box Testing

Unit Testing

Unit testing will be used to test individual functions or methods in isolation to make certain they are performing as intended.

Statement Coverage

Statement coverage testing will be used to verify that every executable line of code is tested at least once.

This will confirm that there is no unreachable or orphaned lines of code within the application.

Branch Coverage

Branch coverage testing will be used to confirm that all True or False conditions are being exercised.

Black-Box Testing

Equivalence Partitioning

Assumptions I made

- A standard employee can only edit their benefits during the open enrollment period unless an HR employee overrides the restriction.
- There are required fields when entering employee or dependent information.
- The benefit rates are determined by plan type, coverage tier, and the benefit period.
- Medical coverage for (employee only) is company-paid, while dental and vision coverage are (employee-paid).
- A duplicate dependent is determined by the same person having the same identifying information (name, DOB and relationship to the employee).

Enrollment Period Rules				
What is Being Tested	Input	Valid?	Expected Result	Related Test Condition and User Stories

Benefits editing during open enrollment	Enrollment Open? (Yes)	Yes	Benefits can be edited	TC2 (US2, US23)
Benefits editing outside of open enrollment	Enrollment Open? (No)	No	Benefits are read-only and message displayed	TC2 (US23)
HR Employee Override outside open enrollment	HR Employee Override	Yes	Changes allowed and logged	TC11 (US7, US23, US27, US30)

Dependent Entry Validation				
What is Being Tested	Input	Valid?	Expected Result	Related Test Condition and User Stories
Relationship provided	Relationship (Child)	Yes	The dependent is allowed	TC3 (US3, US25)
Relationship missing	Relationship (blank)	No	The dependent is not allowed and a validation message shown	TC3 (US25)

Duplicate dependent	Same name, DOB and relationship	No	The duplicate dependent is rejected with an error message	TC3 (US3, US25)
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Boundary Value Analysis

Open Enrollment Period				
What is Being Tested	Invalid	Valid	Invalid	Related Test Condition and User Stories
The ability to editing benefits	Before open enrollment begins	During open enrollment	After open enrollment ends	TC2 (US2, US23)

Session Timeout				
What is Being Tested	Invalid	Valid	Invalid	Related Test Condition and User Stories
Automatic session logout after inactivity	Before activity begins	During active use	After Inactive for more than 5 minutes	TC2 (US2, US23)

Decision Table Testing

Coverage Tier

Conditions

Condition 1	Employee coverage selected
Condition 2	Spouse selected
Condition 3	One child selected
Condition 4	More than one child selected

Actions	
Action 1	Assign Employee Only tier
Action 2	Assign Employee and Spouse tier
Action 3	Assign Employee and Spouse plus One Child tier
Action 4	Assign Employee and Family tier
Action 5	Block selection

Rules		
Rule	Scenario	Actions
Rule 1	Employee only	Action 1
Rule 2	Employee and Spouse	Action 2
Rule 3	Employee and Spouse plus One Child	Action 3
Rule 4	Employee and Multiple Dependents	Action 4

Rule 5	Dependent selected without employee	Action 5
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Conditions	Rules				
	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5
Condition 1	Yes	Yes	Yes	Yes	No
Condition 2	No	Yes	Yes	Yes	Yes
Condition 3	No	No	Yes	No	NA
Condition 4	No	No	No	Yes	NA

Actions	Rules				
	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5
Action 1	Yes	No	No	No	No
Action 2	No	Yes	No	No	No
Action 3	No	No	Yes	No	No
Action 4	No	No	No	Yes	No
Action 5	No	No	No	No	Yes

Test Plan

Tester:	Test Date:
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- **Test Objective #1:** To confirm that employees can view their current benefit selections and dependents
- **Dependencies:** Tester has a compatible Chrome web browser.
- **Tested on:** Google Chrome Version 145.0.7632.76

Step	Procedure Description	Expected Result	Actual Result	Pass/Fail
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1	Login as Emp1	The employee benefits dashboard loads		
2	Navigate to Benefits Summary	The benefits summary page is displayed		
3	View the listed plans	Medical, dental and vision plans which are selected are displayed		
4	Review the dependents covered under plans	Covered dependents are listed correctly		
5	Verify the PPO or HMO label is displayed	The medical plan type clearly identifies whether it is a PPO or an HMO		

Developer Approval Signature X_____

Tester:	Test Date:
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- **Test Objective #2:** Verifying that employees can modify benefits during the open enrollment period
- **Dependencies:** Tester has a compatible Chrome web browser.
- **Tested on:** Google Chrome Version 145.0.7632.76

Step	Procedure Description	Expected Result	Actual Result	Pass/Fail
1	Login as Emp1	The employee benefits dashboard loads		

2	Edit the benefit selections	Editing is benefits is enabled		
3	Change medical PPO to HMO	Medical plan type is updated		
4	Add a dental plan	A dental plan is added		
5	Save the changes	Save is successful		
6	Logout and then Login again	The benefits changes persisted to the new user session		

Developer Approval Signature X_____

Tester:	 Test Date:
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- **Test Objective #3:** Verify adding a new dependent is possible and that duplicate dependents are prevented
- **Dependencies:** Tester has a compatible Chrome web browser.
- **Tested on:** Google Chrome Version 145.0.7632.76

Step	Procedure Description	Expected Result	Actual Result	Pass/Fail
1	Login as Emp1	The employee benefits dashboard loads		
2	Add dependent with required fields	The new dependent is saved		

3	Attempt duplicate entry	The duplicate dependent is blocked		
4	Save	Only one is dependent listed		

Developer Approval Signature X_____

Tester:	Test Date:
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- **Test Objective #4:** Verify dependent coverage blocked without employee coverage
- **Dependencies:** Tester has a compatible Chrome web browser.
- **Tested on:** Google Chrome Version 145.0.7632.76

Step	Procedure Description	Expected Result	Actual Result	Pass/Fail
1	Login as Emp1	The employee benefits dashboard loads		
2	Attempt to add dependents dental without employee dental	The system blocks the selection		
3	Attempt save	Saving prevented		
4	Review the benefits summary	No invalid coverage is shown		

Developer Approval Signature X_____

Tester:	Test Date:
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- **Test Objective #5:** Confirm that automatic tier assignment is occurring
- **Dependencies:** Tester has a compatible Chrome web browser.
- **Tested on:** Google Chrome Version 145.0.7632.76

Step	Procedure Description	Expected Result	Actual Result	Pass/Fail
1	Login as Emp1	The employee benefits dashboard loads		
2	Add spouse and child	Tier (Employee, Spouse and Child) is selected		
3	Add an additional child	The tier is recalculated to the (Employee and Family) plan		

Developer Approval Signature X_____

Tester:	Test Date:
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- **Test Objective #6:** Verify the benefits enrollment submission form is created and validated
- **Dependencies:** Tester has a compatible Chrome web browser.
- **Tested on:** Google Chrome Version 145.0.7632.76

Step	Procedure Description	Expected Result	Actual Result	Pass/Fail
1	Login as Emp1	The employee benefits dashboard loads		

2	Go to the benefits submission form	The benefits have been populated to the form correctly		
3	Remove a required field	A validation indicating a missing required field is triggered		
4	Attempt to submit	Submission of the form is blocked		

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Tester: _____ Test Date: _____

- **Test Objective #7:** To confirm that authentication, authorization, and automatic user session timeout is occurring
- **Dependencies:** Tester has a compatible Chrome web browser.
- **Tested on:** Google Chrome Version 145.0.7632.76

Step	Procedure Description	Expected Result	Actual Result	Pass/Fail
1	Attempt to login using invalid credentials	Access should be denied		
2	Login as Emp1	The user session is started		
3	Attempt to access an HR function	Access should be blocked		

4	Remain idle until the user session is timed out	The user session times out and requires you to relogin		
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Developer Approval Signature X_____

Tester:	Test Date:
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- **Test Objective #8:** Verify that medical PPO and HMO are identified after switching plan types
- **Dependencies:** Tester has a compatible Chrome web browser.
- **Tested on:** Google Chrome Version 145.0.7632.76

Step	Procedure Description	Expected Result	Actual Result	Pass/Fail
1	Login as Emp1	The employee benefits dashboard loads		
2	Change Medical PPO to HMO	The selection is updated		
3	View the benefits summary	Medical HMO is displayed correctly		

Developer Approval Signature X_____

Tester:	Test Date:
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- **Test Objective #9:** To confirm that the benefits summary updates after dependent changes are made
- **Dependencies:** Tester has a compatible Chrome web browser.
- **Tested on:** Google Chrome Version 145.0.7632.76

Step	Procedure Description	Expected Result	Actual Result	Pass/Fail
1	Add new dependent	The benefits summary updates immediately		
2	Remove the dependent you added	The benefits summary updates again		

Developer Approval Signature X_____

Tester:	Test Date:
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- **Test Objective #10:** To verify that if an HR employee changes an effective date it updates immediately, and that audit logging records the changes made
- **Dependencies:** Tester has a compatible Chrome web browser
- **Tested on:** Google Chrome Version 145.0.7632.76

Step	Procedure Description	Expected Result	Actual Result	Pass/Fail
1	Login as HR1	The HR dashboard loads		
2	Update the effective date	The date is saved		
3	View the enrollment summary	The new date is displayed		
4	Check the audit log	The date change recorded was recorded		

Developer Approval Signature X_____

Tester: | **Test Date:**

- **Test Objective #11:** Confirm that an employee record is validated and must be unique
- **Dependencies:** Tester has a compatible Chrome web browser.
- **Tested on:** Google Chrome Version 145.0.7632.76

Step	Procedure Description	Expected Result	Actual Result	Pass/Fail
1	Login as HR1	The HR dashboard loads		
2	Add a new employee and leave a required field blank	Saving blocked and a validation error message is displayed		
3	Enter valid data for the missing required field	Saving the new employee is successful		
4	Attempt to create another new employee using the same duplicate ID	Saving the new employee is rejected and a message explain that the employee ID is already in use and it must be unique		

Developer Approval Signature X_____

Tester: | **Test Date:**

- **Test Objective #12:** Confirm that the rate tables can be maintained and that entries are being validated
- **Dependencies:** Tester has a compatible Chrome web browser.
- **Tested on:** Google Chrome Version 145.0.7632.76

Step	Procedure Description	Expected Result	Actual Result	Pass/Fail
1	Login as HR1	The HR dashboard loads		
2	Select an existing benefit and enter valid new rate	The change is saved successfully		
3	Select an existing benefit and enter a negative rate	The change is rejected and a message displays the error and how to correct it.		
4	Add new period rate for a year that does not currently have a rate assigned to it	The new periods rate does not overwrite any prior period rates		

Developer Approval Signature X_____

Tester:	 Test Date:
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- **Test Objective #13:** Verify that company vs employee paid is correctly being displayed
- **Dependencies:** Tester has a compatible Chrome web browser.
- **Tested on:** Google Chrome Version 145.0.7632.76

Step	Procedure Description	Expected Result	Actual Result	Pass/Fail
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1	As an employee with no spouse and no dependents select Medical PPO, Dental and Vision benefits coverage	The cost is shown correctly		
2	Review labels	Medical PPO is clearly shown as company paid		
3	Review benefits cost totals	The benefits costs are accurately displaying that the employee is being charged for only the dental and vision plans		

Developer Approval Signature X_____

Tester:	Test Date:
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- **Test Objective #14:** Confirm that and HR employee can maintain departments and that the rules regarding the change and removal of a department are being followed
- **Dependencies:** Tester has a compatible Chrome web browser.
- **Tested on:** Google Chrome Version 145.0.7632.76

Step	Procedure Description	Expected Result	Actual Result	Pass/Fail
1	Add a new department	Added department is saved		
2	Assign a nonexistent department to an employee	The change is blocked and a message explains the issue		

3	Delete a department that is currently in use	The deletion is prevented and a message explains the issue		
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Developer Approval Signature X_____

Tester:	Test Date:
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- **Test Objective #15:** Verify that audit logs are accurate, complete and read-only
- **Dependencies:** Tester has a compatible Chrome web browser.
- **Tested on:** Google Chrome Version 145.0.7632.76

Step	Procedure Description	Expected Result	Actual Result	Pass/Fail
1	Make an HR change	The change is made		
2	Open the audit log	It shows who made the change when the change was made and what the change was		
3	Attempt to edit the audit log	The edit is blocked and there is a message explaining that the log is read only and that editing is not allowed		

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Tester: | Test Date:

- **Test Objective #16:** Confirm that role-based user access restrictions are being applied
- **Dependencies:** Tester has a compatible Chrome web browser.
- **Tested on:** Google Chrome Version 145.0.7632.76

Step	Procedure Description	Expected Result	Actual Result	Pass/Fail
1	Login as Emp1	The user session begins		
2	Try to access HR functions	Access is denied		
3	Login as HR1	Access is granted		

Developer Approval Signature X_____

Tester: | Test Date:

- **Test Objective #17:** Confirm that authentication rejects an invalid login attempt
- **Dependencies:** Tester has a compatible Chrome web browser.
- **Tested on:** Google Chrome Version 145.0.7632.76

Step	Procedure Description	Expected Result	Actual Result	Pass/Fail
1	Enter the wrong password	Login is rejected		
2	Enter valid credentials	User session is created		

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Exploratory Testing Sessions

Each testing session will be timeboxed for 60 minutes and will produce testing notes, screenshots, a list of detected defects, and a short session summary report.

Session 1 - Enrollment (Odd Paths)

Testing Guidelines: Try to break the benefits enrollment process by changing plans repeatedly, quickly adding and then removing dependents, navigating back and forward, jumping from one screen to another unrelated screen repeatedly and saving at odd times.

Focus: These actions are meant to test data persistence, the benefits summary refresh under extreme conditions, the timing of tier recalculation, and error handling.

Session 2 - Validation and Error Messaging

Testing Guidelines: Attempt to leave required fields empty, input data in the wrong formats, enter invalid dates, try to enter extremely long entries in text fields, and try to add duplicate dependents.

Focus: These testing actions are to verify the system is producing field validation messages, to highlight any behavior which allowed by the system but was an outlier issue that was never planned for, to test the limits of blocking rules which prevent invalid entries, and to verify the clarity of validation error messages.

Session 3 - Security and Permissions Abuse

Testing Guidelines: Attempt direct navigation to HR function URLs as Emp1, try browser refresh after the session has timed out, attempt the same actions in another browser tab, verify the session lockouts.

Focus: Testing the user-role enforcement consistency, session management, unauthorized action blocking.

Session 4 – Rates and Effective Date Risk Exploration

Testing Guidelines: This test is to stress the rates table by performing edits across multiple periods, trying extreme values so that the limits of what the system will and will not allow are known, verify which rate is applied after the effective date changes have been made.

Focus: Testing benefits period isolation, test expiration, expiring and future rates, calculation rates accuracy after changes are made.

Testing Efforts Estimate

Effort Breakdown		
Activity	Description	Estimated Duration
Testing Plan Finalization	Review requirements, finalize scripts and user story traceability	1 Day
Testing Environment Setup	Verify test data, roles, browser compatibility	1 Day
Execute Critical Tests	Eligibility rules, cost calculations, security, authentication	3 Days
High Priority Tests	Enrollment workflow, validation, summaries, audit checks	2 Days
Medium and Low Priority Tests	Administrative features, department maintenance	1 Day
Exploratory Testing Sessions	4 structured sessions focused on testing edge cases	2 Days
Verifying Defects and Follow Up Regression Testing	Re-run impacted test objectives after fixes	2 Days
Test Reporting	Metrics, defect summaries, final evaluation	1 Day
Total Estimated Test Effort is 13 Days for a Single Tester		

Testing Team Roles and Resources

Roles	
Role	Responsibilities
Test Lead	Owns the test planning, coordination of testing execution, validation of any defects discovered and test reporting
Tester	Executes the scripted test objectives and exploratory testing
Technical Support	Assists with testing environment setup and investigates any testing environment issues
HR Subject Matter Expert	Consultant for business rules and proper enrollment behavior
Project Stakeholder	Reviews testing results and approves application release

Resources Required	
Resource	Purpose
Chrome Browser	Testing platform
Test Environment	A sandboxed instance of the application with seeded employee and rate data
Test Accounts	Role-based validation

Defect Tracking Tool	Logging and tracking issues
Requirements and User Stories	Traceability validation
Test Data Sets	Dependents, plans, and rate configurations

Plan for Incident Management and Reporting

Defect Lifecycle

A potential defect is identified during test execution or an exploratory session.

It is then logged with:

- The defect description
- The steps required to reproduce the defect
- The expected behavior
- The actual behavior
- Screenshots when possible
- A level of severity classification for the defect (critical, high, medium, low)

Next the potential defect is reviewed by the test lead and a developer to determine for certain that a defect does exist and what to do about it.

If the defect is determined to be more than just an incident it is assigned for correction.

After a fix is ready to be deployed, it is retested to make certain the defect has been resolved.

Once the defect passes, retesting with success, the issue is closed.

If the defect persists or does not pass retesting the issue is reopened.

Defect Severity Classification	
Severity	Definition

Critical	Prevents enrollment, causes incorrect cost, or exposes security risk
High	Major functionality broken but workaround exists
Medium	Partial feature issue or validation defect
Low	Cosmetic or usability issue

Key Metrics

Execution Metrics	
Metric	Purpose
Test Case Pass Rate	Measures functional stability
Defect Density	Indicates defect concentration
Critical Defect Count	Determines release risk
Retest Success Rate	Validates fix effectiveness
Requirements Coverage	Ensures all user stories validated

Quality Metrics	
Metric	Purpose
Defect Leakage	Measures escaped defects after testing
Regression Failure Rate	Indicates stability of changes
Validation Error Frequency	Highlights data integrity issues
Security Test Pass Rate	Ensures access control reliability

Risk Analysis

Risk	Impact	Likelihood	Mitigation Method
Incorrect Benefits Cost Calculations	Payroll and financial errors	Medium	Extensive calculation validation and regression testing

Eligibility Rule Failures	Invalid enrollments	Low	Decision-table and white-box validation
Unauthorized Access to HR Functions	Data privacy breach	Low	Role-based testing and security validation
Rate Table Misconfiguration	Incorrect deductions	Medium	Effective-date and boundary testing
Data Integrity Errors	Employee record inaccuracies	Medium	Validation and duplicate detection tests
Session Timeout Failure	Unauthorized workstation access	Low	Security-focused exploratory testing
Late Requirement Changes	Rework and delays	Medium	Maintain traceability and regression suite
Insufficient Test Data	Incomplete validation	Low	Pre-seeded controlled datasets

References

Van Veenendaal, E., Black, R., Graham, D., & Services, R. B. R. B. C. (2019a). Foundations of software testing: ISTQB Certification.